Detection of Fake Hotel Reviews

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Abstract

With the rise of large language models (LLMs), which make text generation cheap and scalable, fake reviews become an increasing concern across industries. We expand on the work of Ignat et al. (2025) by comparing five bag-of-words classifiers to distinguish human-written from AI-generated reviews. We use the English reviews of the MAiDE-up dataset, split into a training set of 1,500 reviews and test set of 500 reviews. Hyperparameters are tuned using 5-fold cross-validated grid search, and model accuracies are compared on the test set using McNemar's test. Naive Bayes achieves the highest accuracy (0.922), while both tree ensembles significantly outperform a single Decision Tree. Feature analysis suggests differences in vocabulary, with AI-generated reviews tending towards more formal language and human reviews towards narrative expressions. These findings indicate that LLM-generated reviews differ from human-written reviews in word use, and that ensembles improve on a single Decision Tree for this task.

Keywords: Fake review detection, AI-generated text, large language models, hotel reviews, text classification, bag-of-words, ensemble methods

1. Introduction

When choosing a hotel, people increasingly rely on online reviews, which can strongly influence their booking decisions. Research shows that negative information tends to weigh more heavily, while positive reviews combined with numerical ratings increase trust and booking intentions (Sparks and Browning, 2011). As reviews can directly affect a hotel's reputation and bookings, fake reviews, also known as opinion spam, are a growing concern for both hotels and customers.

Large language models (LLMs) have improved significantly in recent years, particularly in text generation. Before this, the volume of fake reviews was likely limited by human writing speed, time and energy. By enabling rapid, large-scale review generation, they remove the time and effort constraints of human writing, which makes fake reviews increasingly common (Ignat et al., 2025). Platforms such as Amazon have already reported a surge of ChatGPT-generated fake reviews, which reduce consumer trust in online reviews and harm seller performance (Zhao et al., 2025). Therefore, automated methods are needed to filter out AI-generated reviews.

Prior work has found that AI-generated fake reviews are more comprehensible, less specific, more mechanical and less empathetic than authentic reviews (Zhao et al., 2025). In a recent study, Ignat et al. (2025) collected real hotel reviews from Booking.com and generated fake ones with GPT-4. However, Ignat et al. only compared two models, Gaussian Naive Bayes and Random Forests, to a fine-tuned deep learning model. We extend on this

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prior research and evaluate how effectively simple bag-of-words models can distinguish AI-generated from human-written hotel reviews.

We use the English subset of the MAiDE-up dataset[1] (Ignat et al., 2025) to compare five bag-of-words classifiers: Multinomial Naive Bayes, Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. These models were chosen because they are computationally efficient and interpretable, which allows us to inspect which words drive their predictions. Thereby, we address the following research questions:

1. Does a discriminative linear model (Logistic Regression) outperform a generative one (Multinomial Naive Bayes)?
2. Does combining many trees into an ensemble (Random Forest and Gradient Boosting) improve on a single Decision Tree?
3. Is the Random Forest better at telling fake from genuine reviews when the reviews are positive or when they are negative?
4. Which words are most predictive for each model, and which of them point towards AI-generated or human-written reviews?

Answering these questions shows whether more complex models are worth their cost for this task, and which words distinguish AI-generated reviews from human-written ones.

2. Data

This study investigates the detection of fake hotel reviews by distinguishing between human-written reviews and AI generated reviews. The dataset thus, contains the two classes 0 (human) and 1 (AI). For this study, the dataset was restricted to English-language reviews to ensure linguistic consistency. Further, this restriction facilitates the application of natural language processing techniques and allows the analysis to focus on differences between human and AI-generated reviews rather than on variations based on language-specific characteristics. However, our findings may not generalize to reviews written in other languages. The filtered dataset consists of 2,000 reviews, including 1,000 genuine and 1,000 AI-generated reviews, with ten feature columns and one target variable. First, the two review columns, `Upside_Review` and `Downside_Review`, were combined into a single `Review` column to facilitate subsequent analysis. To obtain an initial overview of the textual characteristics of reviews in both classes, corresponding word clouds were generated (see Appendix B).

The word frequencies show similarities between the two groups: words such as “the”, “and”, and “was” occur frequently in both sources. This can be explained by the grammatical structure and the descriptive nature of hotel reviews. However, slight differences can also be observed for example “hotel” occurring 858 times in the AI-generated reviews and 461 times in the human-written reviews. Overall, the initial data exploration revealed no substantial differences in word frequency.

We therefore conducted a more detailed analysis of punctuation usage (e.g., commas, semicolons, etc.) and the frequency of contractions. The objective was to use these findings to better justify the preprocessing decisions, namely to maintain features that differed

1. https://github.com/MichiganNLP/multilingual_reviews_deception

between the two classes, as they were considered potentially informative for classification. When both classes showed similar patterns, we applied normalization to reduce irrelevant variation and ensure consistent text representation. This follows the recommendations of Hickman et al. (2022), who emphasize that preprocessing decisions should be guided by the research objective and the characteristics of the dataset.

3. Data Preprocessing

Preprocessing transforms the raw reviews into a consistent representation while preserving potentially informative linguistic features. The overall methodology is illustrated in Figure 1.

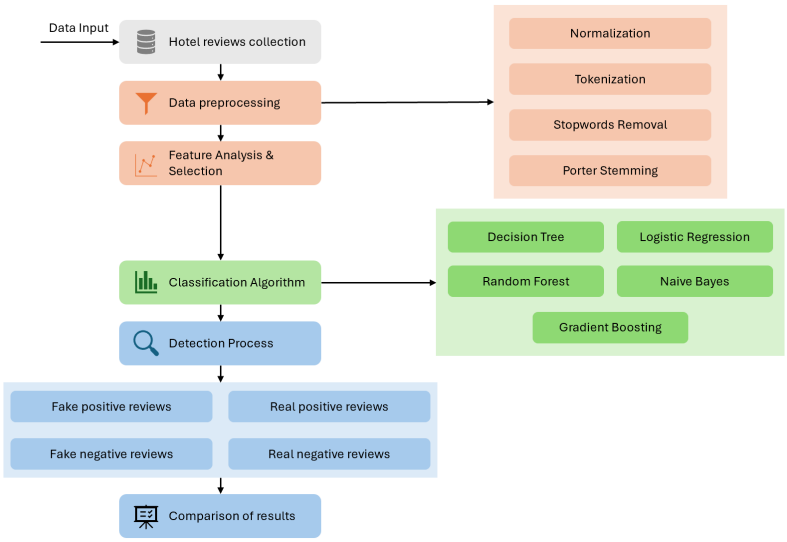


Figure 1: Steps and Techniques used in Sentiment Analysis.

3.1. Normalization

Text normalization is the first step in preprocessing and aims to transform the original text into a more consistent format. In this study, unicode normalization, accent removal, and conversion to lowercase were applied. Additionally, multiple consecutive whitespaces were replaced with a single whitespace. These steps reduce variations and ensure that equivalent terms are represented consistently. This is particularly useful for text classification, as differences in capitalization or character encoding could otherwise lead to identical terms being treated as separate features.

However, normalization requires careful consideration because certain textual characteristics may provide useful information to distinguish human and AI-generated reviews Shah et al. (2023). Therefore, our preprocessing decisions were guided by an exploratory analysis of the dataset to identify differences between the two classes. The analysis shows differences in punctuation and number usage. Human reviews contained more question marks and exclamation marks per 1,000 words than AI-generated reviews (0.63 vs. 0.02 and 5.66

vs. 0.85), whereas AI-generated reviews contained more commas (49.02 vs. 35.06). Additionally, human-written reviews contained substantially more numbers (11.46 vs. 0.83 per 1,000 words). A similar pattern was identified for contractions. Human-written reviews contained 10.22 contractions per 1,000 words, compared with 5.57 in AI-generated reviews. Based on these findings, we decided to keep numbers, punctuation marks, and contractions during text normalization.
This approach aims to balance textual consistency with the preservation of features that may help the model distinguish between the two classes. Nevertheless, the observed differences are descriptive and do not necessarily imply that these features will improve classification performance.

3.2. Tokenization

After normalization, the reviews are split into tokens using whitespace tokenization. Since the dataset only contains English reviews, in which spaces generally separate words, this is a simple and computationally efficient choice; subword tokenizers such as SentencePiece (Kudo and Richardson, 2018) are mainly needed for languages without clear word boundaries. However, punctuation may remain attached to words and contractions may not be consistently split into separate tokens.

3.3. Stop-Word Removal

After tokenization, we remove stop words such as *the*, *a* and *and* using the English stop-word list of the Natural Language Toolkit (NLTK). Since these function words showed little discriminative value in the initial analysis (Section 2), removing them shifts the focus to content words about rooms, staff, service and location. However, common words can also carry stylistic information, so their removal may discard some useful signal.

3.4. Stemming

The final step is stemming, which reduces words to a common stem so that related word forms are grouped and the vocabulary becomes smaller. We use the Porter stemmer, a widely used algorithm designed for English (Willett, 2006). The resulting stems are not always valid words (e.g., *servic*) and unrelated words may occasionally share a stem, but the narrow hotel-review domain limits this risk.

4. Experimental Setup

After preprocessing, the dataset is split into stratified training and test sets to preserve the 50/50 class distribution. No separate validation set is needed, because hyperparameters are tuned with 5-fold cross-validation (see Table 6 in Appendix A). 5-fold cross-validation was chosen to balance robust evaluation against dataset size and computational time. All models use a bag-of-words representation, which represents reviews through word counts. While computationally efficient, this approach does not capture word order.

1. Multinomial Naive Bayes As a standard model for text classification, it counts word frequencies per class. By beginning with this baseline, we can evaluate whether increased model complexity yields improvements in accuracy. We used the chi-squared test to select the top k features, which assists in removing features that are less informative for prediction. Additionally, the minimum document frequency (min_df) was tuned via grid search to remove sparse terms. However, the selected k was all features, indicating that feature selection did not improve performance.

2. Logistic Regression This is another common and computationally efficient baseline, as it calculates a weight for each term within the bag-of-words representation, making the model highly interpretable. To prevent overfitting given the limited sample size, L1 regularization was applied, which removes uninformative terms by setting their weights to zero. The regularization strength C (where $C = 1/\lambda$) was tuned over a range of 0.01 to 100.

3. Decision Tree This model was chosen for its interpretability, allowing for a clear analysis of the factors driving its predictions. Furthermore, since a Random Forest model is also implemented, a single Decision tree serves as a necessary baseline to determine if an ensemble approach improves performance, as documented in the literature (Breiman, 2001). The Decision Tree Classifier was implemented using the scikit-learn library with the default Gini setting. However, since the default cost-complexity pruning parameter α (ccp_alpha) is 0, we performed a grid search to optimize this hyperparameter. Without tuning α the tree can become overly complex and lead to overfitting. The final model for testing was refit on the entire training set once the optimal α was found.

4. Random Forest (Breiman, 2001) A single Decision Tree has a high variance, since small changes in the training data can lead to a very different tree. Random Forest reduces variance by combining trees trained on bootstrap samples of the training data. At each split, only a random subset of features is considered, which makes the trees less correlated. We tuned min_df, the number of features considered at each split (max_features) and the number of trees (n_estimators). The trees were grown to full depth, as is common for Random Forest, because averaging over many trees counteracts the overfitting of the individual trees. The selected max_features was $\log_2 p$, which with a vocabulary of $p = 4443$ words means that each split only considers 12 randomly chosen words, resulting in strongly decorrelated trees. The largest number of trees in the grid (500) was selected, so adding more trees might improve the performance slightly further.

5. Gradient Boosting (Friedman, 2001) Whereas Random Forest builds independent trees, Gradient Boosting builds trees sequentially, where each new tree is fitted to the errors (the gradient of the log-loss) of the trees before it. Thereby, Gradient Boosting mainly reduces bias by adding up many weak learners, typically shallow trees. Including both ensemble methods allows us to compare two different approaches of combining decision trees. We tuned the learning rate, which scales the contribution of each tree, the number of trees and the maximum depth of each tree. Because the trees have to be trained one after another, Gradient Boosting cannot be parallelized over the trees like Random Forest. To limit the computation time, we used a smaller grid and kept min_df at its default value. The selected maximum depth was 1, meaning that each tree is a decision stump that splits

on a single word. This was combined with the highest learning rate (0.2) and the largest number of trees (300) in the grid.

In addition to accuracy, precision, recall and F1-score, we used McNemar's test (Dietterich, 1998), defined in Equation (1), to assess whether differences in accuracy between models are significant. We compare Naive Bayes and Logistic Regression as generative and discriminative models, respectively (Ng and Jordan, 2001), to assess which approach is more effective for text classification. Random Forest and Gradient Boosting are compared against the Decision Tree baseline to evaluate whether ensembles improve accuracy over a single tree.

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}} \tag{1}$$

5. Results

5.1. Performance

Multinomial Naive Bayes achieved the highest test accuracy (0.922) and F1-score (0.925) of all models, compared to 0.896 and 0.894 for Logistic Regression. However, their cross-validation accuracies are nearly identical (0.899 and 0.901), suggesting that the gap on the test set may be specific to this split (see Section 5.2). Naive Bayes has a higher recall (0.960) but lower precision (0.892), so it misses fewer AI-generated reviews but also incorrectly labels more human reviews as AI. Logistic Regression shows the opposite pattern with a higher precision (0.909) and lower recall (0.880). In comparison to the other models, the Decision Tree performs worst on all metrics. Random Forest achieves the highest cross-validation accuracy of all models (0.905) and a test accuracy of 0.910, second only to Naive Bayes. It also has the highest precision of all models (0.956), so it rarely labels human reviews as AI-generated. Gradient Boosting reaches a slightly lower accuracy (0.888) and F1-score (0.885). Both ensembles show the same pattern as Logistic Regression, with a lower recall (0.860 and 0.864) than precision (0.956 and 0.908), meaning they miss more AI-generated reviews than they wrongly flag human ones.

Table 1: Cross-validation accuracy on the training set and test set performance of the final models.

Model	**CV accuracy**	**Accuracy**	**Precision**	**Recall**	**F1**
Naive Bayes	0.8993	0.9220	0.8922	0.9600	0.9249
Logistic Regression	0.9013	0.8960	0.9091	0.8800	0.8943
Decision Tree	0.7820	0.7800	0.7966	0.7520	0.7737
Random Forest	0.9053	0.9100	0.9556	0.8600	0.9053
Gradient Boosting	0.8927	0.8880	0.9076	0.8640	0.8852

5.2. Statistical Tests

The McNemar's test results for Multinomial Naive Bayes and Logistic Regression are detailed in Table 2. As shown, both models correctly classify the vast majority of reviews (421 instances). Logistic regression misclassifies 40 reviews that Naive Bayes correctly identifies, while the reverse occurs in 27 instances. Although Naive Bayes makes fewer unique errors, McNemar's test found no significant accuracy difference ($p = 0.143 > 0.01$).

Table 2: McNemar contingency tables on the test set, with McNemar's χ^2 statistic (Equation (1)) and p-value below each table. LR = Logistic Regression, NB = Naive Bayes, RF = Random Forest, GB = Gradient Boosting, DT = Decision Tree.

(a) LR vs. NB

	NB correct	NB wrong
LR correct	421	27
LR wrong	40	12

$\chi^2 = 2.149$, $p = 0.143$

(b) RF vs. DT

	DT correct	DT wrong
RF correct	371	84
RF wrong	19	26

$\chi^2 = 39.77$, $p < 0.001$

(c) GB vs. DT

	DT correct	DT wrong
GB correct	370	74
GB wrong	20	36

$\chi^2 = 29.88$, $p < 0.001$

We used McNemar's tests further to assess whether Random Forest and Gradient Boosting significantly outperform the Decision Tree baseline. For Random Forest versus the Decision Tree (Table 2b), Random Forest corrected 84 errors but introduced 19 new ones. McNemar's test confirmed a significant improvement ($\chi^2 = 39.77$, $p < 0.001$), rejecting the null hypothesis at the 0.01 significance level. Similarly, Gradient Boosting corrected 74 errors made by the Decision Tree and introduced 20 new ones. McNemar's test confirmed a significant improvement ($\chi^2 = 29.88$, $p < 0.001$) over the single Decision Tree.

5.3. Feature Importance

The most important features identified by each model are summarized in Table 3.

Table 3: Top 5 features of the linear models. For Naive Bayes the score is the log-probability ratio $\log P(w \mid \text{AI}) - \log P(w \mid \text{human})$; for Logistic Regression it is the L1 coefficient. Positive scores point towards AI-generated, negative towards human-written reviews.

Naive Bayes				**Logistic Regression**			
AI		Human		AI		Human	
notch	3.2481	told	−3.7142	lack	2.5434	construct	−2.3260
bucharest	3.1528	said	−3.5806	delight	2.2315	book	−2.1892
downsid	3.0097	10	−3.5065	heart	2.2170	like	−1.5718
furthermore	2.8871	com	−3.5065	limit	2.0602	fine	−1.5499
desired	2.8871	etc	−3.3395	proxim	1.9040	door	−1.4353

The top five features for Multinomial Naive Bayes have similar scores. This suggests that they contribute nearly equally to the model's predictions. Some features appear as word stems due to Porter stemming (see Section 3.4). These results suggest that each word was approximately $e^{2.89} \approx 18$ to $e^{3.25} \approx 26$ times more likely to appear in AI-generated reviews. With 22 to 51 occurrences each, these terms likely contributed significantly to Naive Bayes classification. Their formal characteristics, such as "desired", suggest that GPT-4-generated reviews tend to use more formal language than human-written reviews. "Notch" likely stems from "top-notch". "Furthermore" is a distinct outlier in the list of words, since it is a transition word. This suggests that GPT-4 uses structural connectors more frequently than humans.

The top five features identified by Logistic Regression do not overlap with those of Naive Bayes. This difference can be attributed to Naive Bayes modeling the generative probability of individual words, whereas Logistic Regression models the class boundary. Additionally, *lack* was identified as a sign of AI-generation, suggesting that AI may focus more on missing services, whereas human reviews may do so less frequently. The features associated with AI generation appear to be more formal and polished in nature. Furthermore, AI tends to describe the appearance of locations in polished terms, whereas humans focus more on actions ("told," "said," "book") and concrete details ("construct," "door"), consistent with Zhao et al. (2025). This may reflect a difference between AI-generated descriptions based on likely impressions and human accounts of lived experiences, as suggested by Ignat et al. (2025). The number *10* among the human features is consistent with the exploratory analysis in Section 3.1, where human-written reviews contained substantially more numbers per 1,000 words than AI-generated reviews (11.46 vs. 0.83). The outlier"com" likely originates from "booking.com" and appears as a separate token due to tokenization. This resulted in Naive Bayes picking up on a platform-specific artifact of the human reviews, whereas AI does not mention the booking platforms.

The Decision Tree's top five features ranked by Gini importance, show no overlap with the linear models. However, unlike the previous models, the AI-associated words are mainly about the hotel itself, likely because the tree favours common words that split many reviews.

For example, *hotel* appears in 60% of the AI-generated training reviews but only in 29% of the human-written ones, which is consistent with the word frequency analysis in Section 2. *Central* is rarer but more one-sided, appearing in 16% of AI-generated and 3% of human-written reviews. The only exception is *good*, which is associated with human reviews (28% of human-written vs. 14% of AI-generated reviews). Although descriptive words are mostly associated with AI in the other models, *good* points towards human reviews. This could be because AI uses more polished language (e.g., *delight*, *top-notch*) to appear convincing, whereas humans may use simpler, more neutral adjectives like *good* (and *fine* for Logistic Regression) to express satisfaction.

Random Forest and Gradient Boosting share exactly the same top five features, only in a different order, and all of them point towards AI-generated reviews. Three of these (*hotel*, *servic* and *staff*) are also among the top features of the Decision Tree, suggesting that the ensembles rely on the same hotel-related vocabulary as the single tree. In contrast to the Decision Tree, both ensembles rank *lack* (in 14% of AI-generated vs. 1% of human-written reviews) among their top features. *Lack* is also the top AI feature of Logistic Regression, making it the only word shared between a tree-based and a linear model. The importances of the ensembles are spread more evenly across the top words than those of the Decision Tree, since each word's importance is averaged over many trees. For Gradient Boosting, the selected tree depth (max_depth) of 1 means that each tree is a single split on one word, so the model is an additive combination of individual word effects without interactions between words.

Table 4: Top 5 features of the tree-based models, ranked by Gini importance. Direction is the class with the larger share of training reviews containing the word.

Decision Tree			**Random Forest**			**Gradient Boosting**		
hotel	0.1332	AI	servic	0.0133	AI	servic	0.0863	AI
servic	0.1123	AI	staff	0.0128	AI	staff	0.0801	AI
good	0.0557	Human	room	0.0126	AI	hotel	0.0650	AI
central	0.0529	AI	hotel	0.0113	AI	room	0.0561	AI
staff	0.0476	AI	lack	0.0087	AI	lack	0.0454	AI

5.4. Performance by Review Polarity

To examine whether Random Forest performs differently across sentiment polarities, we evaluated its test performance separately for positive and negative reviews. As shown in Table 5, negative reviews achieved slightly higher accuracy (0.9252), precision (0.9746), and F1-score (0.9237) than positive reviews (0.8943, 0.9346, and 0.8850, respectively). However, a chi-square test found no statistically significant difference in accuracy $\chi^2 = 1.1031$ ($p = 0.2936 > 0.01$). Thus, we fail to reject the null hypothesis, suggesting that Random Forest's detection performance does not differ significantly between positive and negative reviews.

Table 5: Random Forest performance by review polarity (positive vs. negative) and contingency table for accuracy ($\chi^2 = 1.103$, $p = 0.2936$).

Polarity	Accuracy	Precision	Recall	F1	Correct	Incorrect
Negative	0.9252	0.9746	0.8779	0.9237	235	19
Positive	0.8943	0.9346	0.8403	0.8850	220	26

6. Limitations and Future Work

First, this study was restricted to English hotel reviews. Although this allowed us to apply a consistent preprocessing pipeline, it limits the generalizability of our findings to other linguistic contexts. As online review platforms serve users worldwide, the expansion of AI-generated review detection to multiple languages represents an important direction for future research. However, differences in grammar, syntax, and writing conventions may affect the accuracy of the detection models.

Second, the study focused exclusively on hotel reviews. However, the characteristics of AI-generated reviews may vary across domains, such as e-commerce, restaurants, and other service industries. Future research could therefore investigate whether the patterns identified in this study also occur in other types of reviews. Domain-specific models could be developed to account for differences in vocabulary, writing style, and review content.

Furthermore, the classification results may depend on the preprocessing techniques and model configurations. A limitation of all models is the bag-of-words representation, which obscures the analysis of word order. Future research could investigate the use of n-grams to capture recurring word combinations, alternative normalization techniques, and different tokenization methods. In addition, different classification models, including deep neural networks, could be explored to evaluate their suitability for AI-generated review detection.

7. Conclusions

In this paper, we compared five bag-of-words classifiers for distinguishing human-written from GPT-4-generated hotel reviews in the English part of the MAiDE-up dataset. First, the generative Multinomial Naive Bayes achieved the highest test accuracy (0.922), but McNemar's test showed no significant difference from the discriminative Logistic Regression (0.896, $p = 0.143$), and their cross-validation accuracies were nearly identical. Second, both tree ensembles significantly outperformed the single Decision Tree ($p < 0.001$): Random Forest raised the accuracy from 0.780 to 0.910 and Gradient Boosting to 0.888. Third, Random Forest was slightly more accurate on negative than on positive reviews (0.925 vs. 0.894), but this difference was not significant ($p = 0.294$).

Feature analysis shows that AI-generated reviews tend to use more formal and generic language, whereas human reviews contain more narrative expressions and numbers. Overall, the results show that simple models can perform well, while more complex models do not always lead to better results.

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AI Statement

Laura van der Hoef I used Claude to review my code, suggest readability improvements and help solve bugs. For example, it suggested how to format the log statements so the output was easier to read and interpret. I also used Claude to help me with styling in Latex, such as setting up the tables in the results section and formatting the text correctly. Claude was used for finding spelling & grammar mistakes and providing suggestions to fix them.

Theresa Schludi I used ChatGPT to assist with debugging code and to create visualizations using matplotlib. Additionally, ChatGPT was used to support Latex formatting, particularly for including references and figures in the report.

Mateusz Hibner I used Claude Opus 5.5 to explain errors, debug, and clean up my code. I also used it to help with Latex formatting, such as creating tables.

Appendix A. Hyperparameter Tuning

Table 6: Hyperparameter grids searched with 5-fold cross-validation and the selected values. Parameter names follow scikit-learn; min_df not listed means the default of 1 was used.

Model	**Hyperparameter**	**Grid**	**Selected**
Naive Bayes	min_df	$\{1, 2, 3, 5, 10\}$	3
	k	$\{50, 100, 200, 300, 500, \text{all}\}$	all
Logistic Regression	min_df	$\{1, 2, 3, 5, 10\}$	1
	C	15 log-spaced in $[0.01, 100]$	1
Decision Tree	ccp_alpha	all α on the pruning path	0.0017
Random Forest	min_df	$\{1, 2, 5\}$	1
	max_features	$\{\sqrt{p}, \log_2 p, 0.05p, 0.1p\}$	$\log_2 p$
	n_estimators	$\{100, 300, 500\}$	500
Gradient Boosting	learning_rate	$\{0.05, 0.1, 0.2\}$	0.2
	n_estimators	$\{100, 300\}$	300
	max_depth	$\{1, 3, 5\}$	1

Appendix B. Word Clouds

Word clouds provide an initial overview of the most frequent words in human-written and AI-generated reviews.

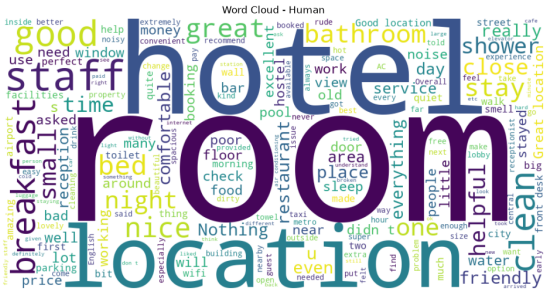


Figure 2: Word cloud of human-written hotel reviews.

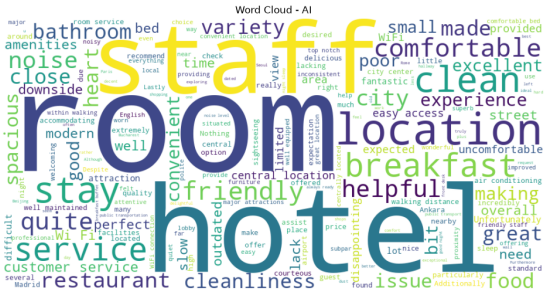


Figure 3: Word cloud of AI-generated hotel reviews.